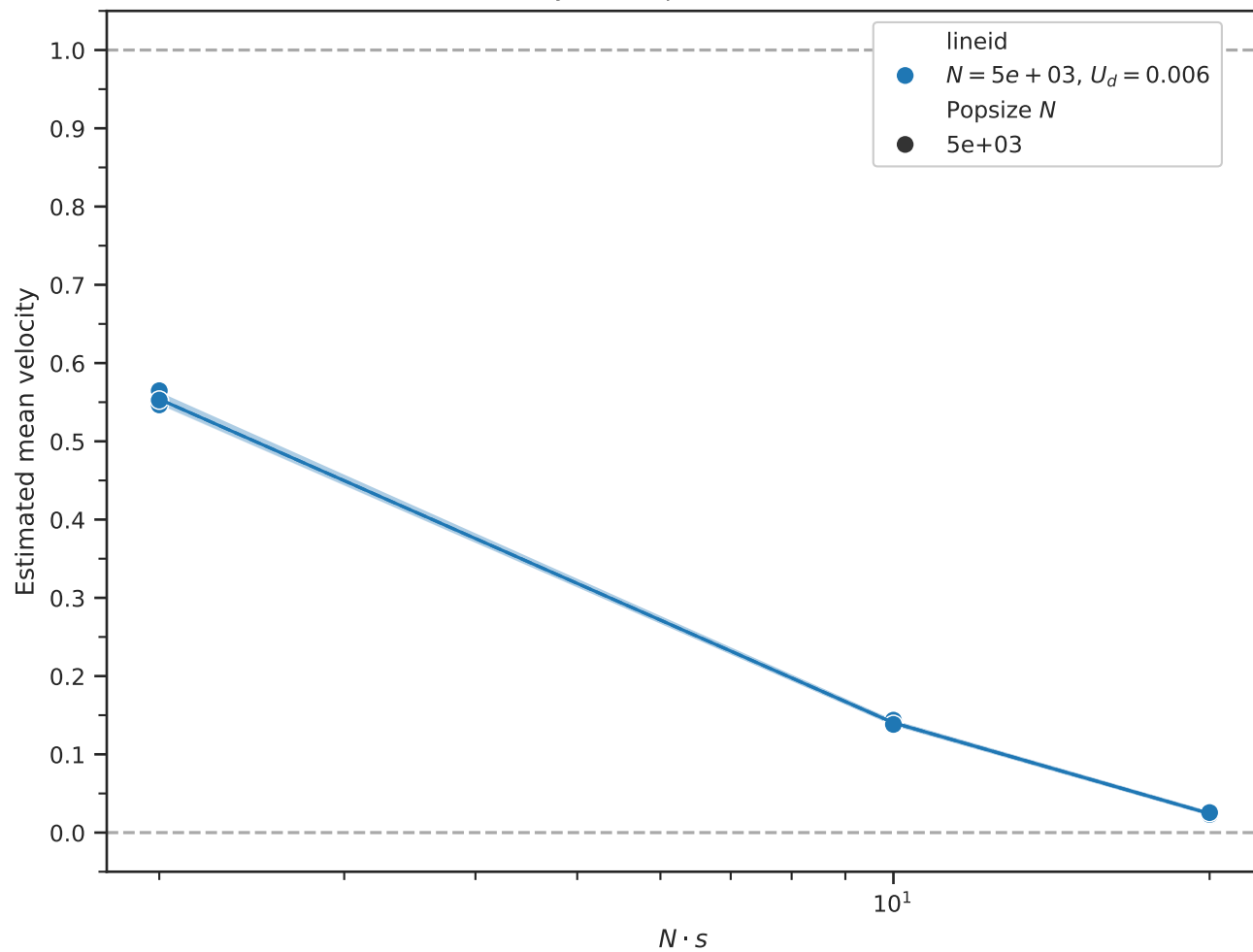
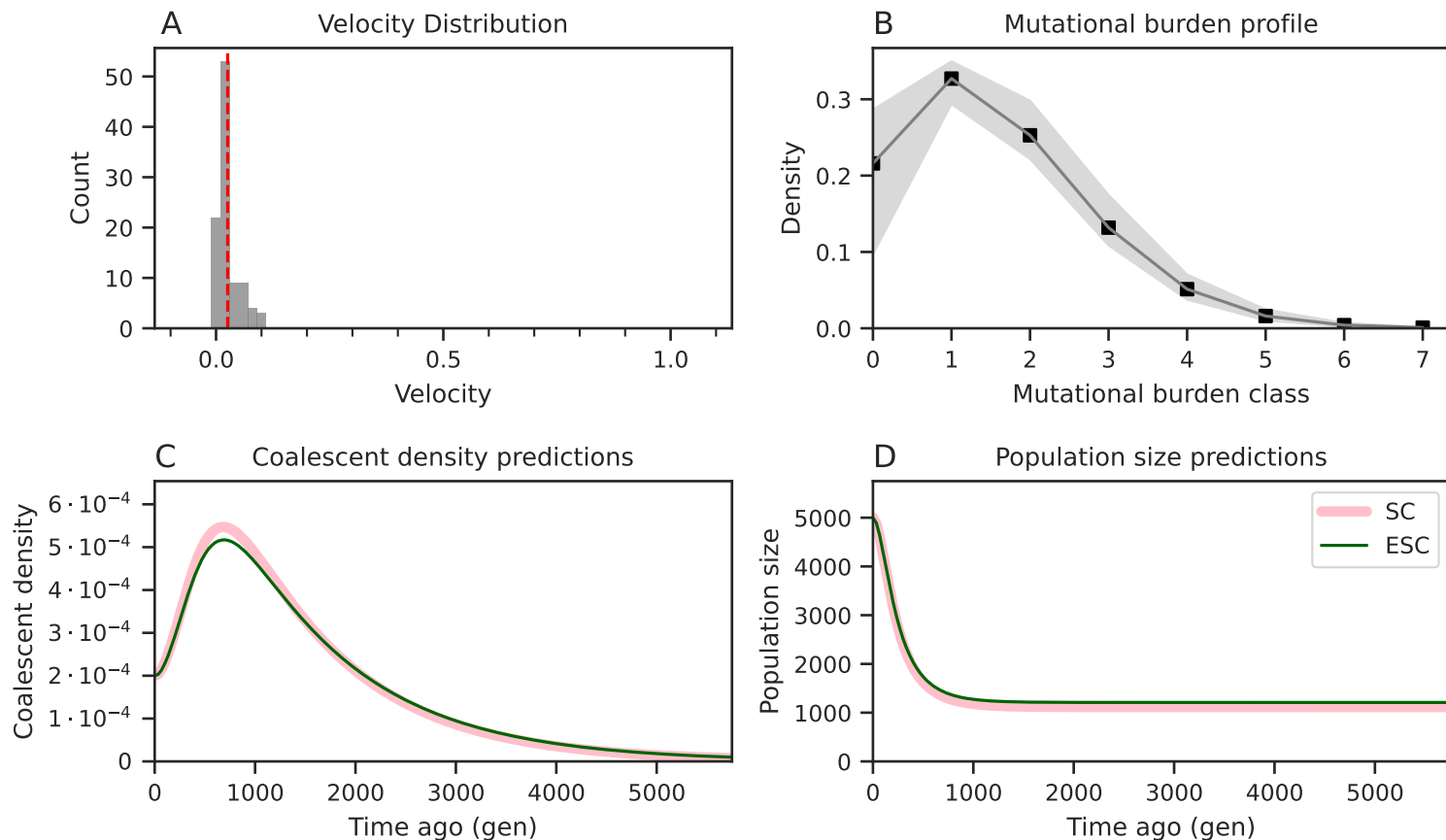


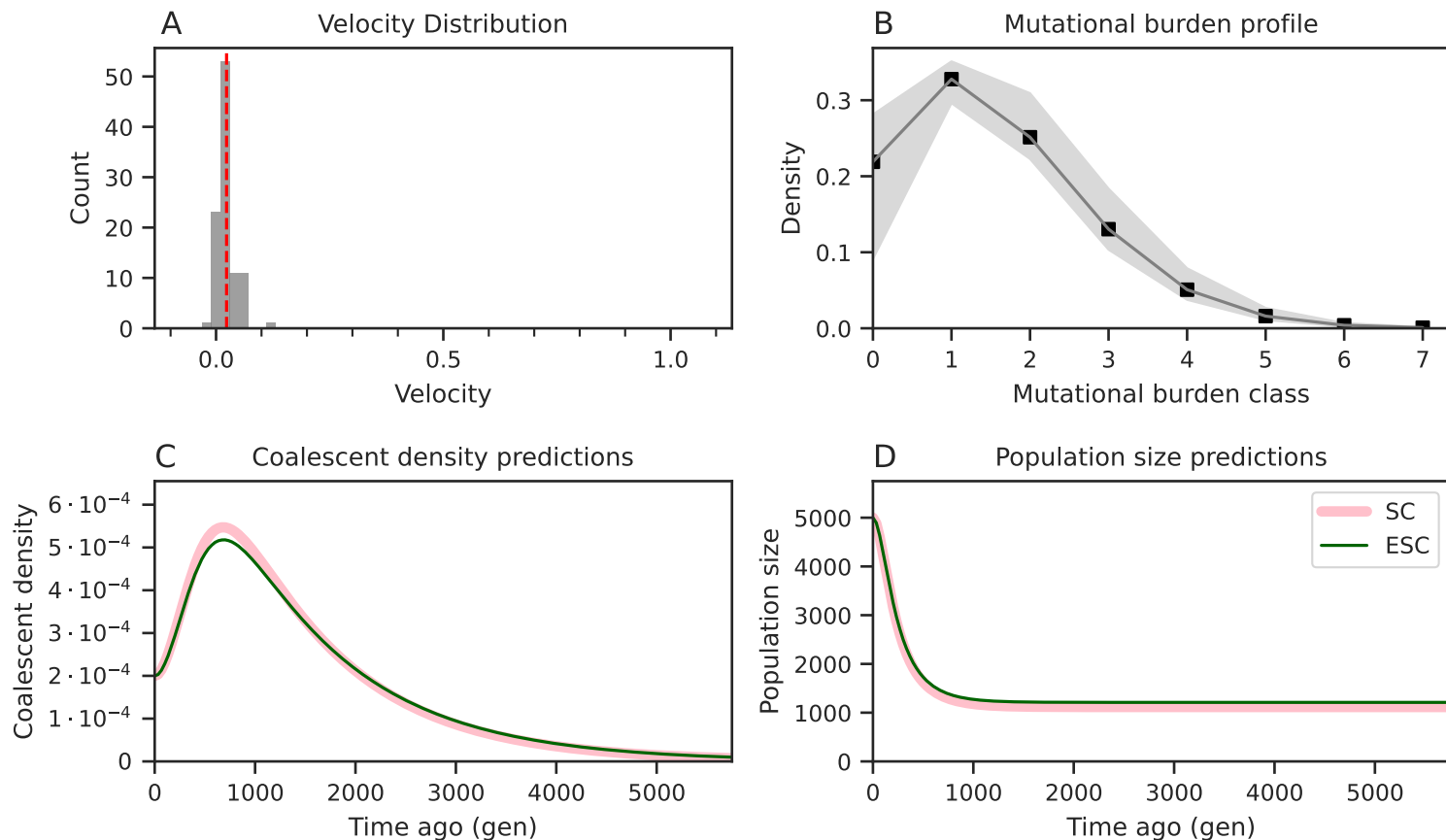
Mean velocity across parameter combinations



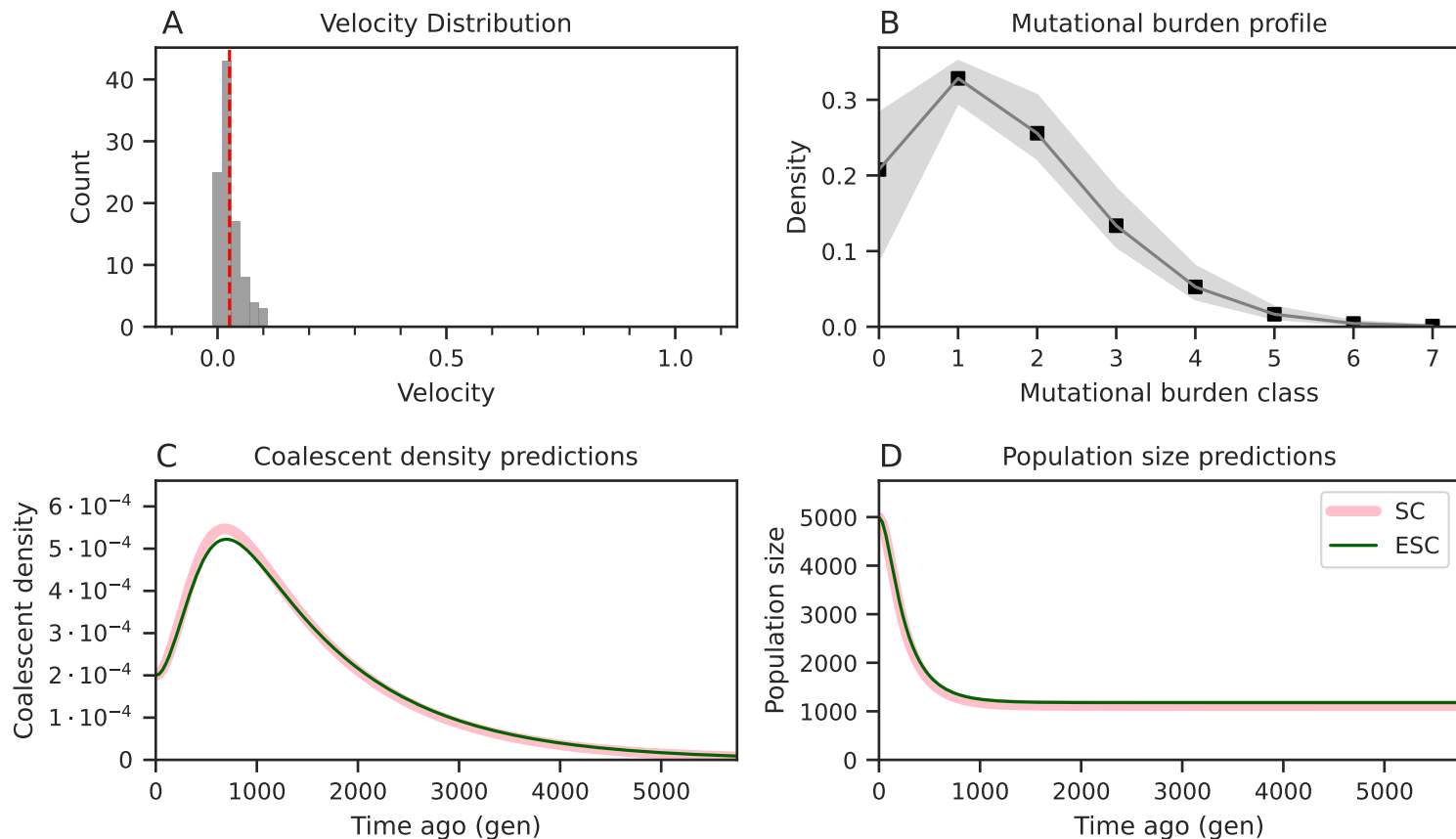
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.002$, $\phi = 4.46$
Mean Velocity= 0.0256 ± 0.0232 , $n = 100$ simulations



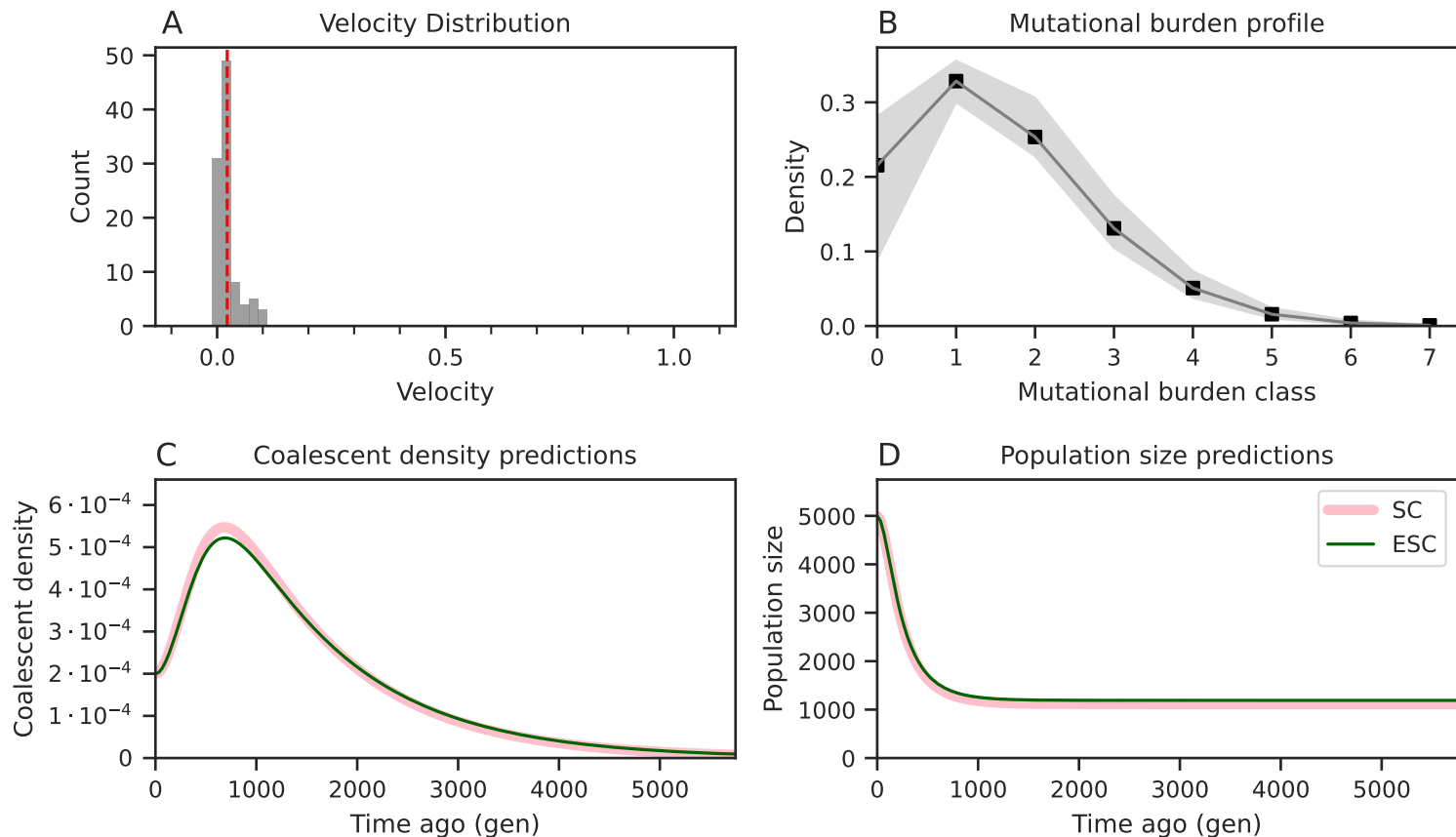
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 4.46$
Mean Velocity= 0.0231 ± 0.0207 , $n = 100$ simulations



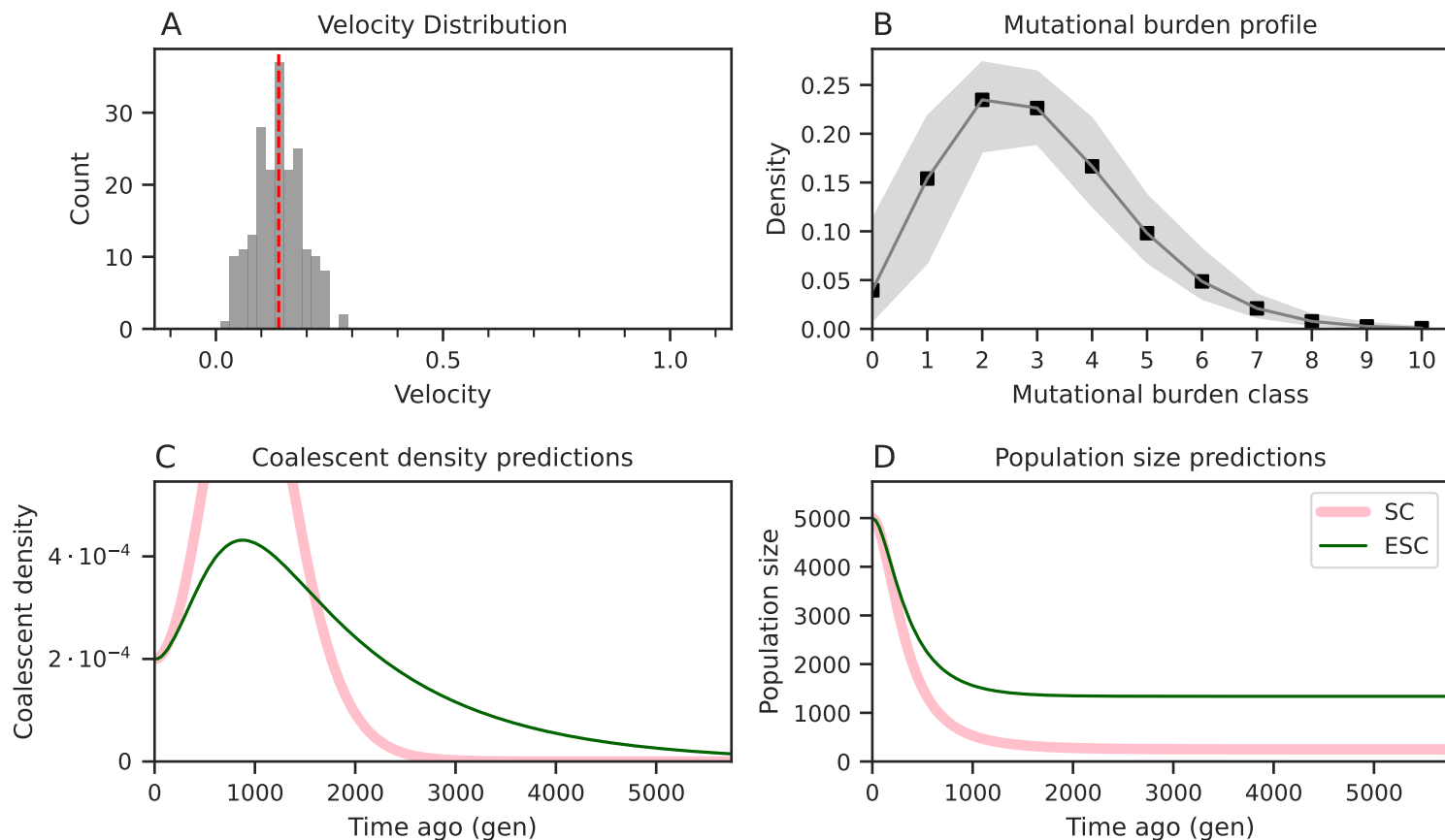
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0262 ± 0.0226 , $n = 100$ simulations



$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity = 0.0222 ± 0.0239 , $n = 100$ simulations

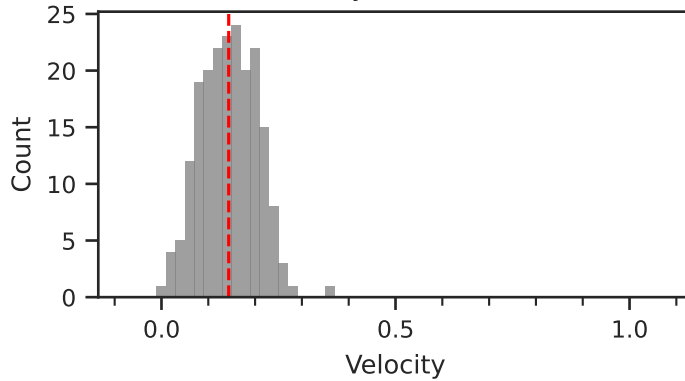


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 0.498$
Mean Velocity= 0.1382 ± 0.0533 , $n = 200$ simulations

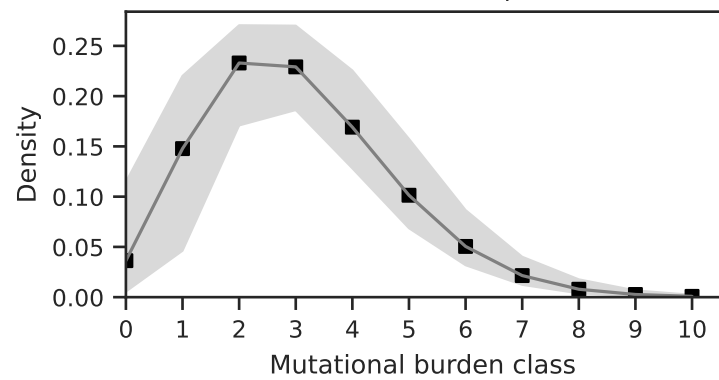


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity = 0.1437 ± 0.0596 , $n = 200$ simulations

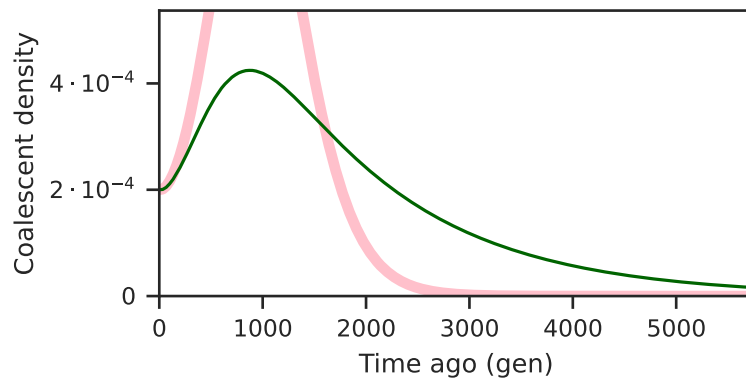
A Velocity Distribution



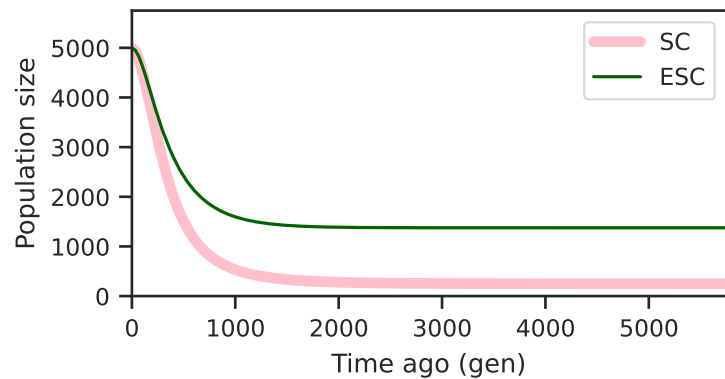
B Mutational burden profile



C Coalescent density predictions

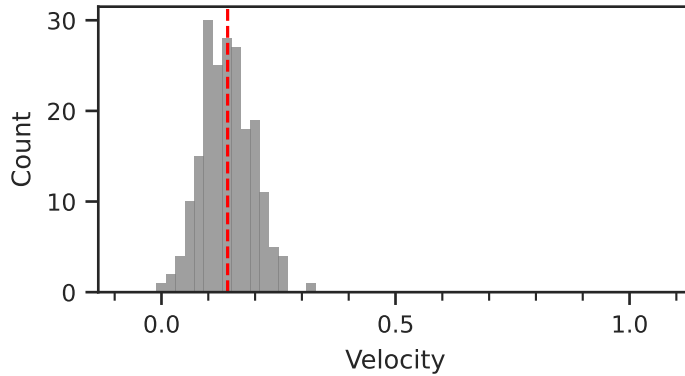


D Population size predictions

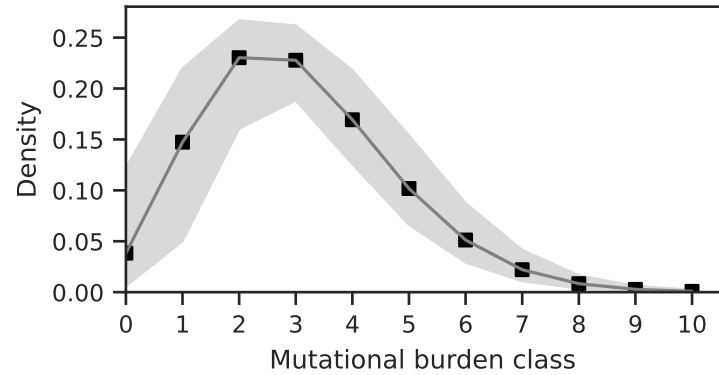


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1413 ± 0.0536 , $n = 200$ simulations

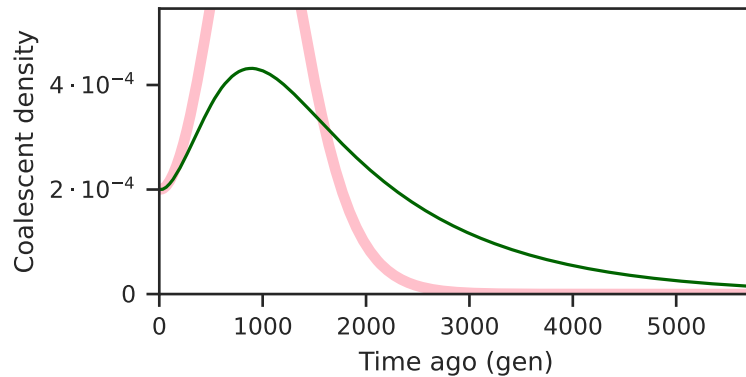
A Velocity Distribution



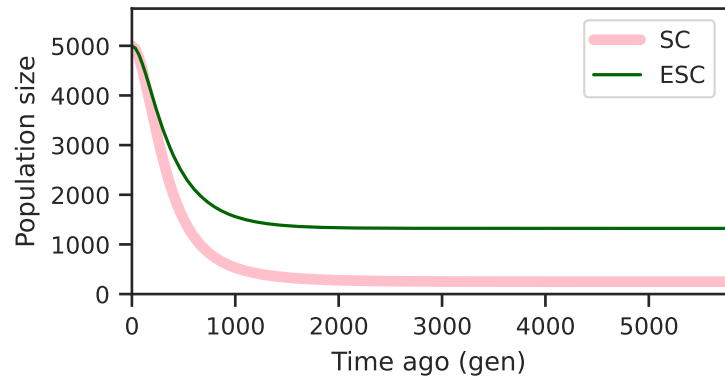
B Mutational burden profile



C Coalescent density predictions

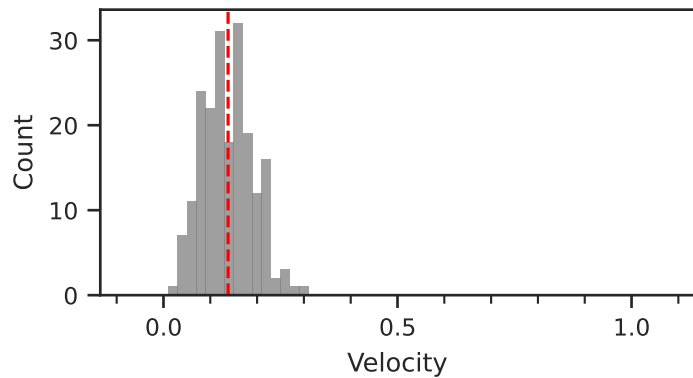


D Population size predictions

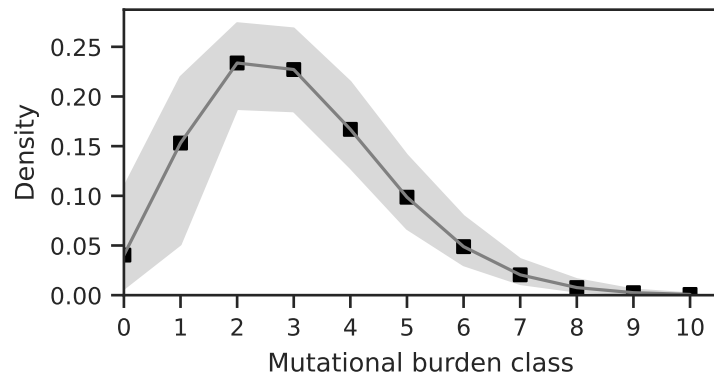


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1380 ± 0.0538 , $n = 200$ simulations

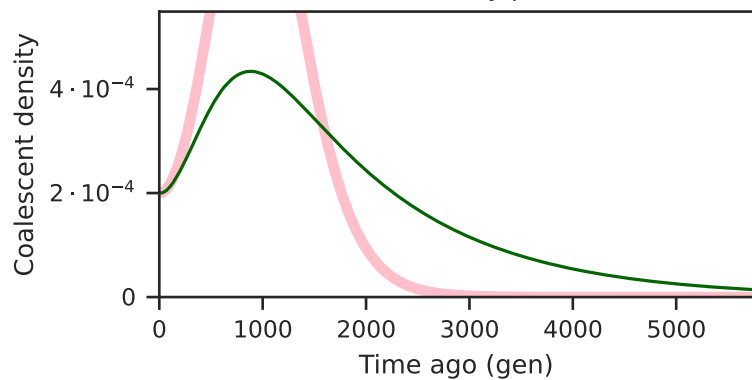
A Velocity Distribution



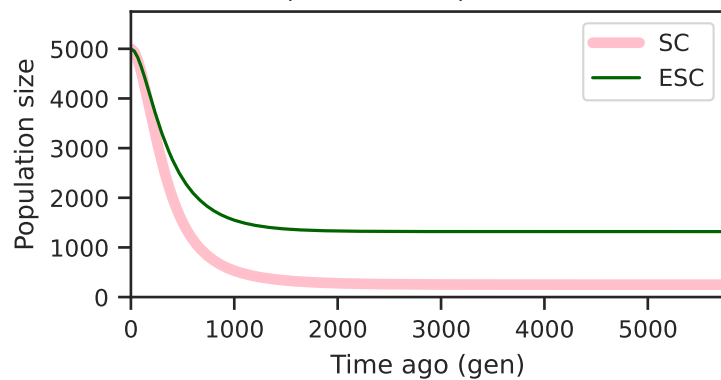
B Mutational burden profile



C Coalescent density predictions

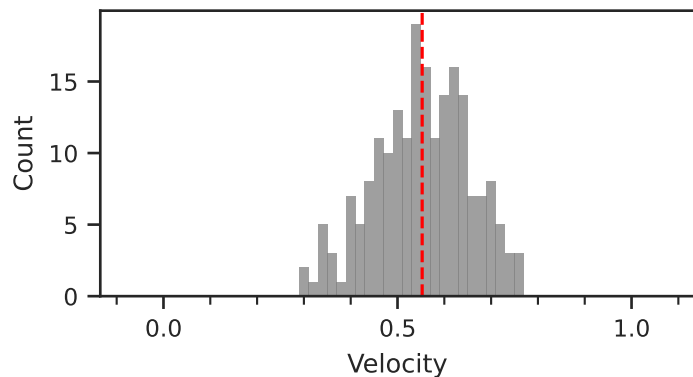


D Population size predictions

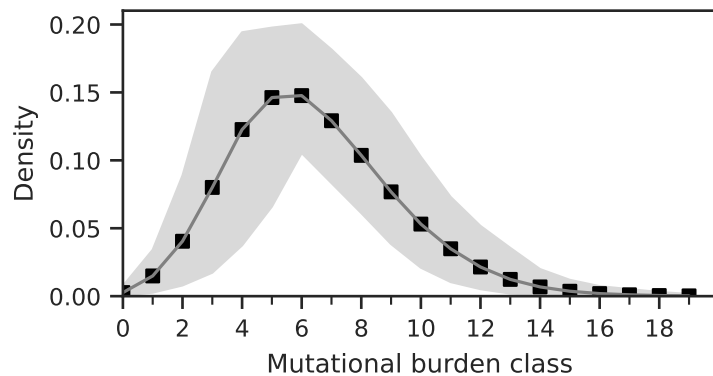


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5527 ± 0.1010 , $n = 200$ simulations

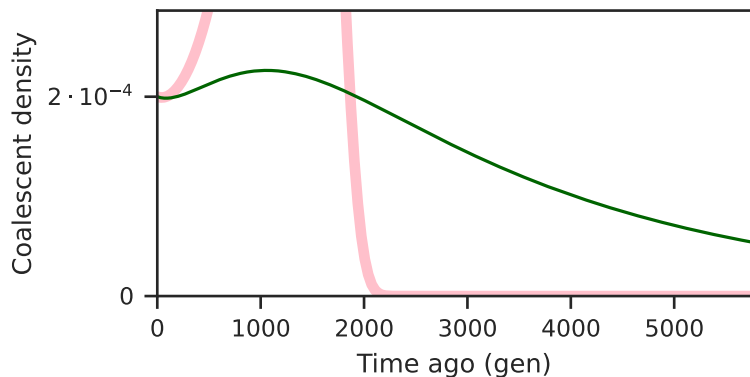
A Velocity Distribution



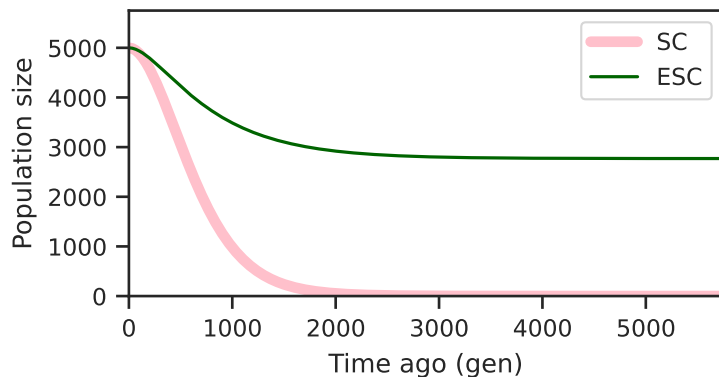
B Mutational burden profile



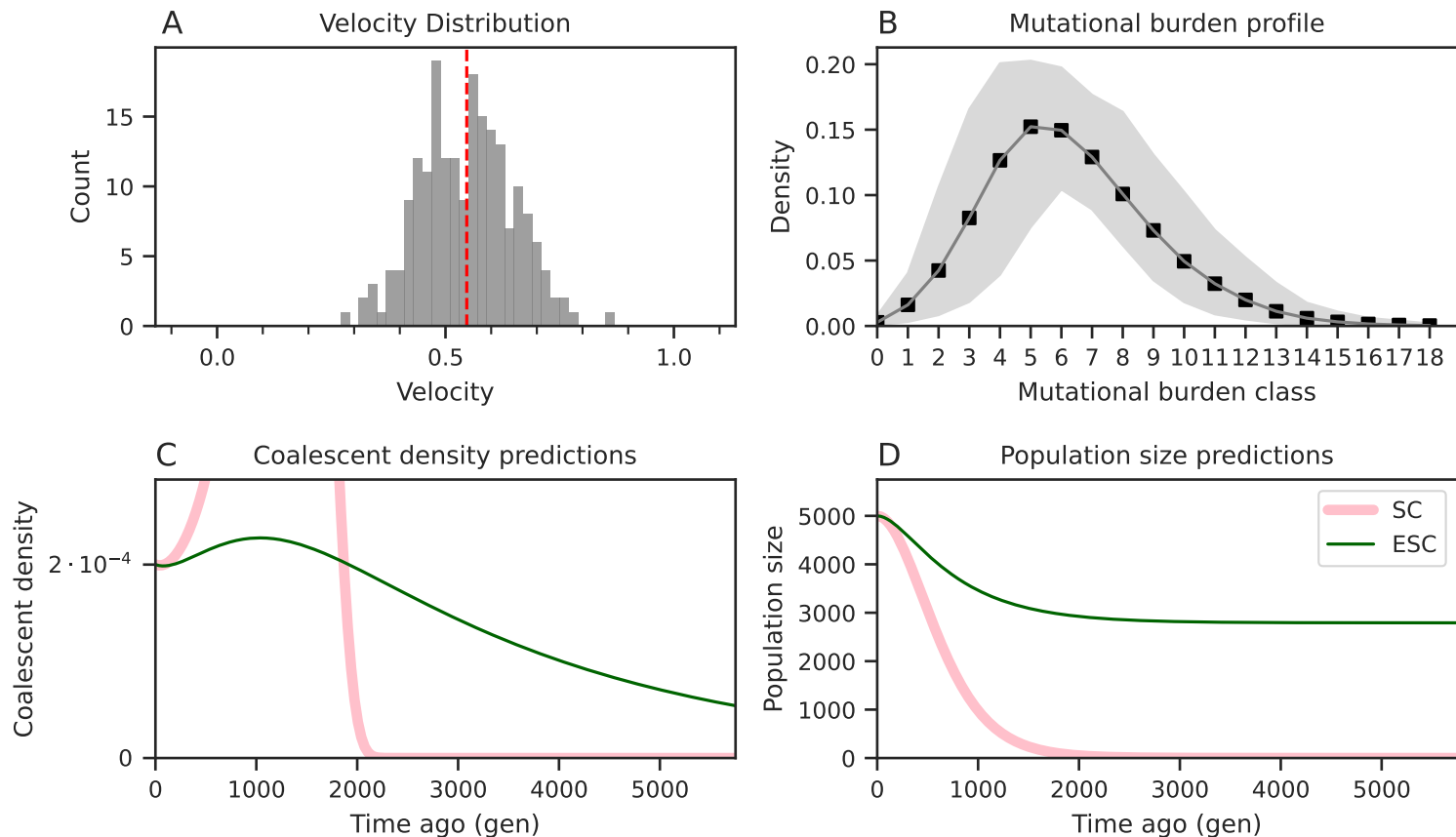
C Coalescent density predictions



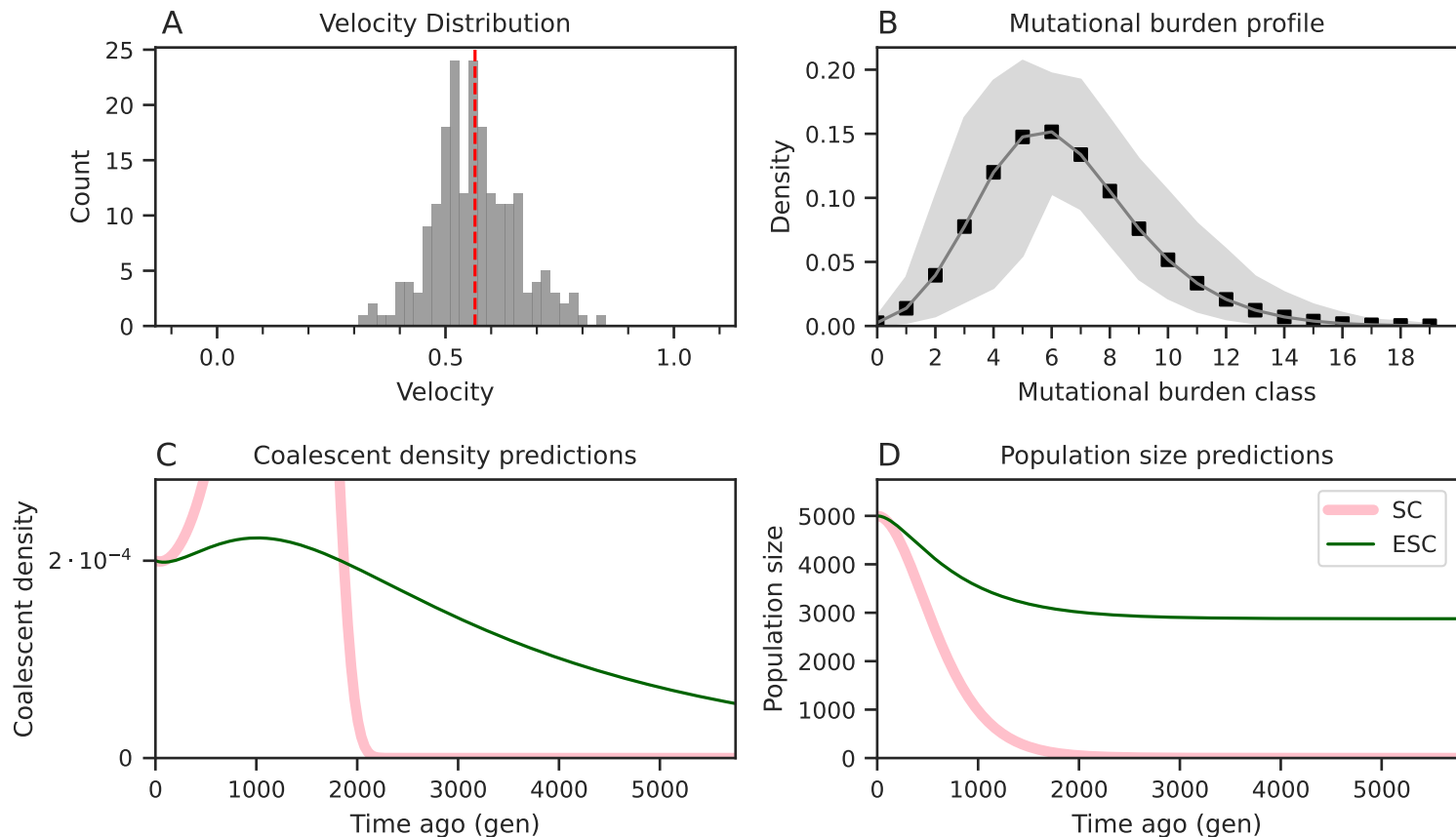
D Population size predictions



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5465 ± 0.1016 , $n = 200$ simulations



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5647 ± 0.0923 , $n = 200$ simulations



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5517 ± 0.1036 , $n = 200$ simulations

